In this, R is integral domain.

Def. A non-zero, non-unit element $r \in R$ is called irreducible if:

if $r = a \cdot b$ for some $a, b \in R$, then either a or b is unit.

In other words, $(r) \subseteq (a) \Rightarrow (a) = (r)$ or $(a) = R$. (That is, (r) is maximal principal ideal.

Prop. Given non-zero, non-unit $r \in R$,

r is irreducible if and only if (r) is maximal principal ideal.

Proof. Suppose that r is an irreducible element. Assume that

$$(r) \subseteq (m) \subseteq R \quad \text{for some } m \in R.$$

Since $(r) \subseteq (m) \Leftrightarrow m \mid r \Leftrightarrow r = mx$ for some $x \in R$,
either m or x must be unit, by the irreducibility.
If m is unit, then given $a \in R$, $a = a \cdot 1 = a \cdot m^{-1} \cdot m \in (m)$, so $(m) = R$.
If x is unit, then $(r) = (mx) = (m)(x) = (m)R = (m)$.
Therefore, (r) is the largest principal ideal.
Conversely, assume that r is reducible. Then, for some non-zero non-unit $x, y \in R$, $r = xy$.
Then, $(r) \subsetneq (x)$ because: if $(r) = (x)$, then $x \in (xy)$, $x = xyz$ for some $z \in R$,
so $x \cdot (1 - yz) = 0$. Since R is integral domain, $yz = 1$, so y is unit, it contradict.
And, $(x) \subsetneq R$, because x is not unit. So, if r is reducible, it is not maximal.

Def. A non-zero, non-unit element $p \in R$ is called prime if:

if $p \mid ab$ for some $a, b \in R$, then either $p \mid a$ or $p \mid b$.

In other words, $ab \in (p) \Rightarrow a \in (p)$ or $b \in (p)$. (That is, (p) is prime ideal.)

Prop. In an integral domain, every prime element is irreducible.